UNIVERSITY OF WESTMINSTER

5COSC024W Server-Side Web Development

2021-2022 TUTORIAL 04 – BASKET – MODEL ANSWER

TASK 01| Create the BASKET page to capture id and quantity of selected product

1) Go to your **H:\homteq** folder and, in the same folder, copy and paste your **template** file and rename the copy to **basket.php.**

2) Open **basket.php** in your code editor, change the name of the page (i.e. the variable **$pagename**) to "Smart Basket".

3) Right at the top under the PHP opening tag, add the following line to connect to the database.

```
include("db.php");
```

4) Locate the line of code that displays the name of the page using the <h4> tag.
Right under this line, create 2 local variables `$newprodid` and `$reququantity` and capture the id of the selected product and the numbers of selected items as passed from the previous page using the POST method. *(Hint: use* `$_POST`*).*
Then display these 2 values for debugging purposes. Use the pseudocode below as the basis for your code.

PSEUDOCODE

```
//capture the ID of selected product using the POST method and the $_POST superglobal variable
//and store it in a new local variable called $newprodid

//capture the required quantity of selected product using the POST method and $_POST superglobal variable
//and store it in a new local variable called $reququantity

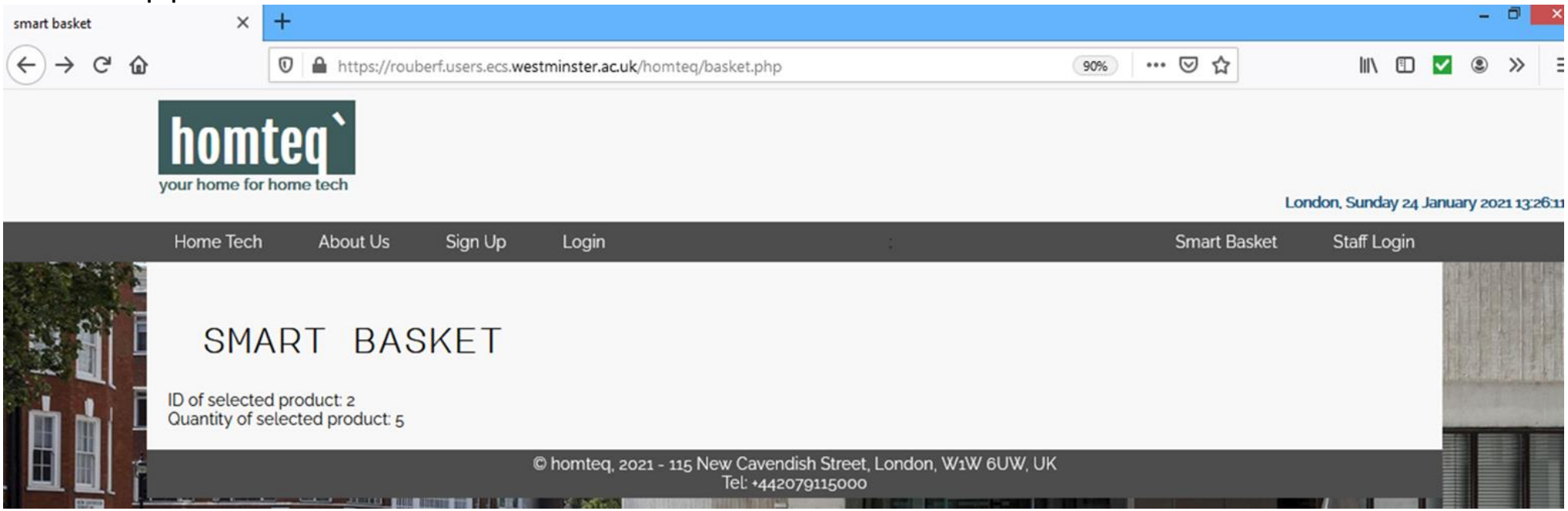
//Display id of selected product
//Display quantity of selected product
```

Hint: adapt the code in Tutorial 03, Task 01, Point 4) and use $_POST instead

5) Upload the **basket.php** file using WinSCP. Then access your INDEX page and select a product, chose a number of items using the drop-down menu and click "ADD TO BASKET".

You should be able to see a display similar to this:

basket.php



The display should show the id of the selected product and the number of items required for this selected product.

Model Answer

```
//capture the ID of selected product using the POST method and the $_POST superglobal variable
//and store it in a new local variable called $newprodid
$newprodid=$_POST['h_prodid'];

//capture the required quantity of selected product using the POST method and $_POST superglobal variable
//and store it in a new local variable called $reququantity
$reququantity=$_POST['p_quantity'];

//Display id of selected product
echo "<p>Id of selected product: ".$newprodid;
//Display quantity of selected product
echo "<br>Quantity of selected product: ".$reququantity;
```

TASK 02| Update the BASKET page to store details of the product reservation in memory

1) Right at the top of **basket.php**, right under the PHP opening tag add the following instruction to initiate the session `session_start();`

2) Right under the code you wrote in Task 01 to display the product id and the quantity of products (which you can later comment out), enter the following code:

```
//create a new cell in the  basket session array. Index this cell with the new product id.
//Inside the cell store the required product quantity
$_SESSION['basket'][$newprodid]=$reququantity;
echo "<p>1 item added";
```

This code updates the session array. It stores the product id as the index of the cell and the required number of products as the content of the cell of the array.

3) Access your INDEX page, select a product, chose a number of items using the drop-down menu and click “ADD TO BASKET”. You should see this:

basket.php



“1 item added” indicates that a new product is added to the basket and thus both the Product Id and the Required Quantity have been stored in the session array.

TASK 03| Update the BASKET page so that it can also be accessed without adding anything onto it

What happens if you access the basket from another page without adding a new product? For instance, go to the Index page and click on the "My Basket" link in the navigation bar. Should the page still display "1 Item added to the basket"? This needs to be addressed.

1) Update the code you have written so far on basket.php to reflect the following pseudocode.
Add the code that reflects the sections highlighted in yellow (you have written the rest already). Make sure you indent the code properly as shown here.

PSEUDOCODE

```
//if the posted ID of the new product is set i.e. if the user is adding a new product into the basket

      //capture the ID of selected product using the POST method and the $_POST superglobal variable
      //and store it in a new local variable called $newprodid

      //capture the required quantity of selected product using the POST method and $_POST superglobal variable
      //and store it in a new local variable called $reququantity

      //Display id of selected product                ← Not required any more, comment this code out
      //Display quantity of selected product          ← Not required any more, comment this code out

      //create a new cell in the basket session array. Index this cell with the new product id.
      //Inside the cell store the required product quantity

      //Display "1 item added to the basket " message

//else
      //Display "Basket unchanged " message
```

*Hint: use the PHP instruction **ISSET** applied to the Superglobal variable $_POST*

Model Answer

```
//if the posted ID of the new product is set i.e. if the user is adding a new product into the basket
if (isset($_POST['h_prodid']))
{
	//capture the ID of selected product using the POST method and the $_POST superglobal variable
	//and store it in a new local variable called $newprodid
	$newprodid=$_POST['h_prodid'];

	//capture the required quantity of selected product using the POST method and $_POST superglobal variable
	//and store it in a new local variable called $reququantity
	$reququantity=$_POST['p_quantity'];

	//Display id of selected product
	//echo "<p>Id of selected product: ".$newprodid;
	//Display quantity of selected product
	//echo "<br>Quantity of selected product: ".$reququantity;

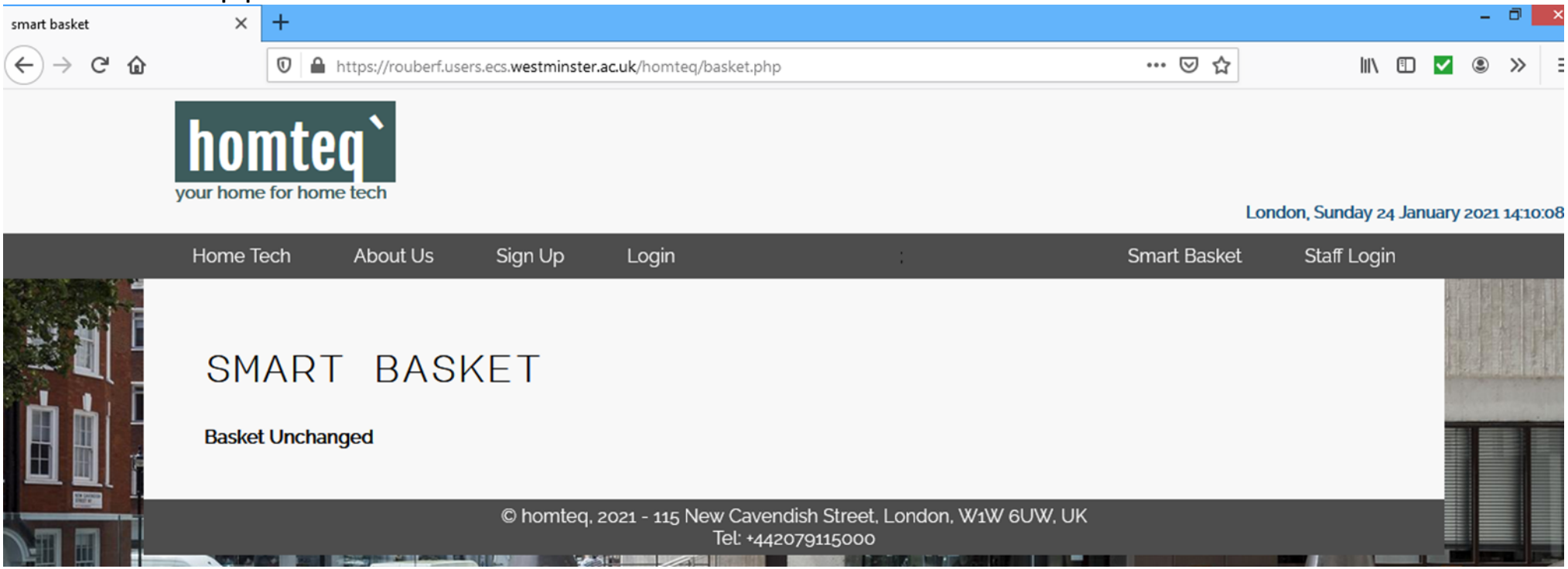
	//create a new cell in the basket session array. Index this cell with the new product id.
	//Inside the cell store the required product quantity
	$_SESSION['basket'][$newprodid]=$reququantity;

	//Display "1 item added to the basket " message
	echo "<p>1 item added";
}
//else
//Display "Current basket unchanged " message
else
{
	echo "<p>Basket unchanged";
}
```

2) Access your INDEX page and click on the “Smart Basket” link on the navigation bar to view the basket without adding anything onto it.

You should be able to see a display similar to this:

basket.php



"Basket unchanged" indicates that the basket is just accessed without a product being added onto it and that thus the session array is unchanged.

TASK 04| Update the BASKET page so that it can display the content of the basket session stored in memory

1) Right underneath the code you have written so far on **basket.php** (i.e., underneath the display of "Basket Unchanged")**,** write code to read from the session array and display the content of the basket so that it reflects the pseudocode shown below.

PSEUDOCODE

```
//Create a variable $total and initialize it to zero

//Create a HTML table with a header to display the content of the shopping basket
//i.e. the product name, the price, the selected quantity and the subtotal

//if the session array $_SESSION['basket'] is set
{
      //loop through the basket session array for each data item inside the session using a foreach loop
      //to split the session array between the index and the content of the cell
      //for each iteration of the loop
      //store the id in a local variable $index & store the required quantity into a local variable $value
      {
            //SQL query to retrieve from Product table details of selected product for which id matches $index
            //execute query and create array of records $arrayp
            //create a new HTML table row
            //display product name using array of records $arrayp
            //display product price using the array of records $arrayp
            //display selected quantity of product retrieved from the cell of session array and now in $value
            //calculate subtotal, store it in a local variable $subtotal and display it
            //increase total by adding the subtotal to the current total
      }
      // Display total
}
//else
{
      //Display empty basket message
}
```

Hint: to create the loop through the session array use `foreach($_SESSION['basket'] as $index => $value)`

For each iteration of the loop, this splits the session array between an index and a value.
It extracts the session index value (here the product id) and stores it in the $index variable.
It extracts the session content value (here the number of required product) and stores it in the $value variable.

Model Answer

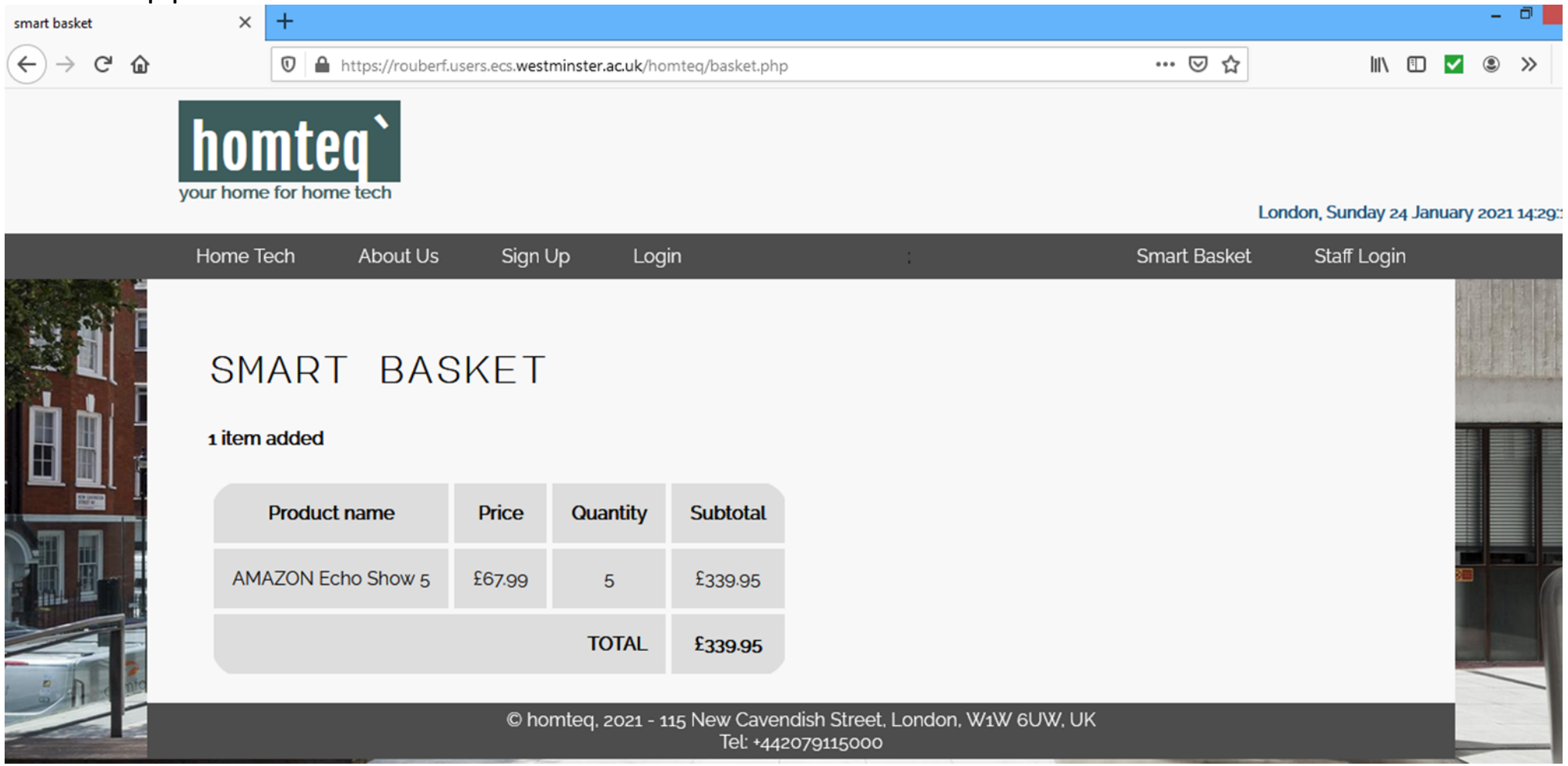
```
$total= 0;    //Create a variable $total and initialize it to zero
//Create HTML table with header to display the content of the basket: prod name, price, selected quantity and subtotal
echo "<p><table id='baskettable'>";
echo "<tr>";
echo "<th>Product Name</th><th>Price</th><th>Quantity</th><th>Subtotal</th>";
echo "</tr>";
//if the session array $_SESSION['basket'] is set
if (isset($_SESSION['basket']))
{
    //loop through the basket session array for each data item inside the session using a foreach loop
    //to split the session array between the index and the content of the cell
    //for each iteration of the loop
    //store the id in a local variable $index & store the required quantity into a local variable $value
    foreach($_SESSION['basket'] as $index => $value)
    {
            //SQL query to retrieve from Product table details of selected product for which id matches $index
            //execute query and create array of records $arrayp
            $SQL="select prodId, prodName, prodPrice from Product where prodId=".$index;
            $exeSQL=mysqli_query($conn, $SQL) or die (mysql_error());
            $arrayp=mysqli_fetch_array($exeSQL);
            echo "<tr>";
            //display product name & product price using array of records $arrayp
            echo "<td>".$arrayp['prodName']."</td>";
            echo "<td>&pound".number_format($arrayp['prodPrice'],2)."</td>";
            // display selected quantity of product retrieved from the cell of session array and now in $value
            echo "<td style='text-align:center;'>".$value."</td>";
            //calculate subtotal, store it in a local variable $subtotal and display it
            $subtotal=$arrayp['prodPrice'] * $value;
            echo "<td>&pound".number_format($subtotal,2)."</td>";
            echo "</tr>";
            //increase total by adding the subtotal to the current total
            $total=$total+$subtotal;
    }
}
//else display empty basket message
else
{
       echo "<p>Empty basket";
}

// Display total
echo "<tr>";
echo "<td colspan=3>TOTAL</td>";
echo "<td>&pound".number_format($total,2)."</td>";
echo "</tr>";
echo "</table>";
```

2) Now access your INDEX page, select a product, chose a number of items using the drop-down menu and click “ADD TO BASKET”.

You should be able to see a display similar to this:

basket.php



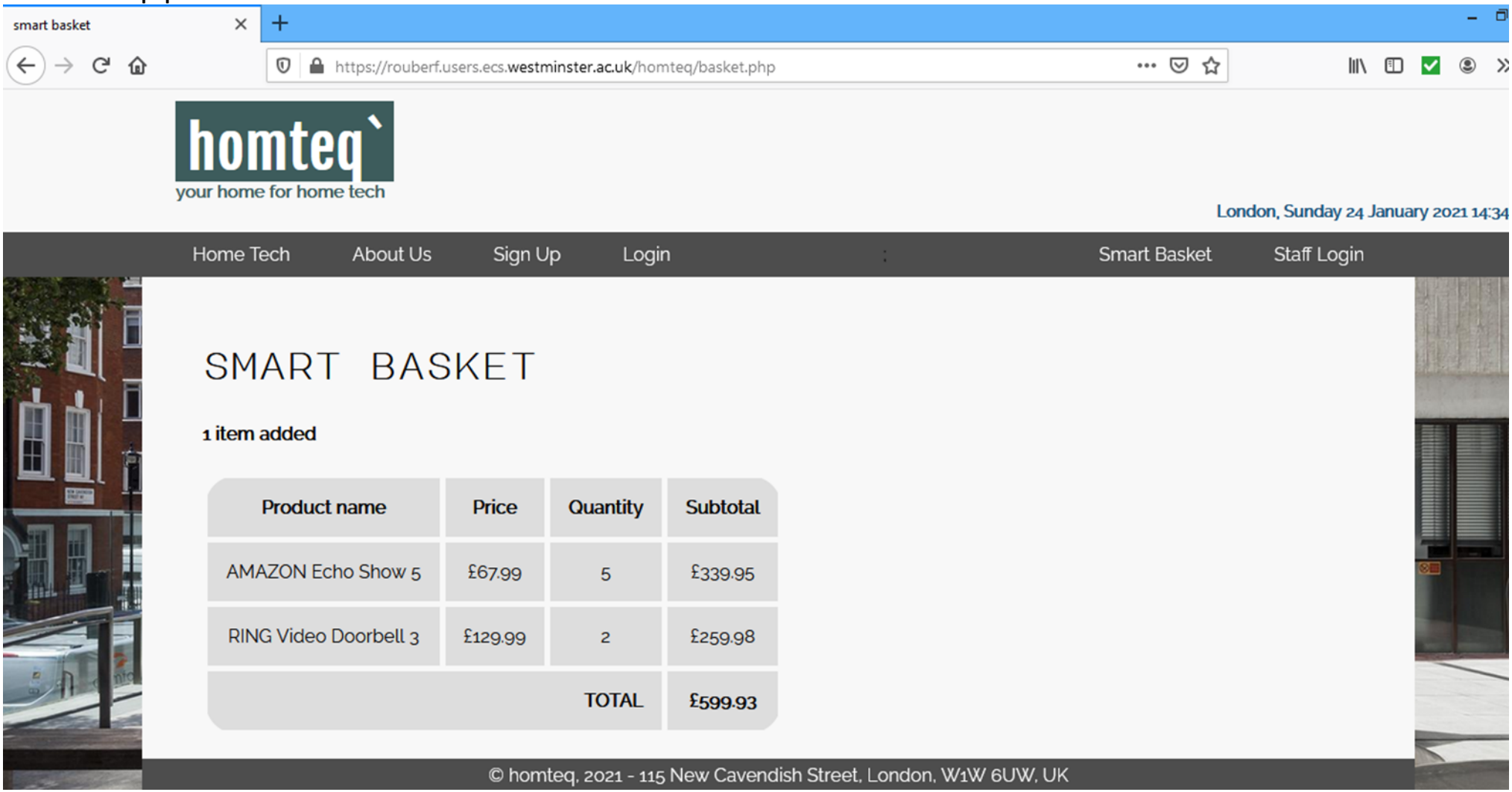
The basket has been uploaded; one item has been added.

The details of the products selected by the user are retrieved and displayed in the basket.

3) Then access your INDEX page, select another product, chose a number of items using the drop-down menu and click “ADD TO BASKET”.

You should be able to see a display similar to this:

basket.php



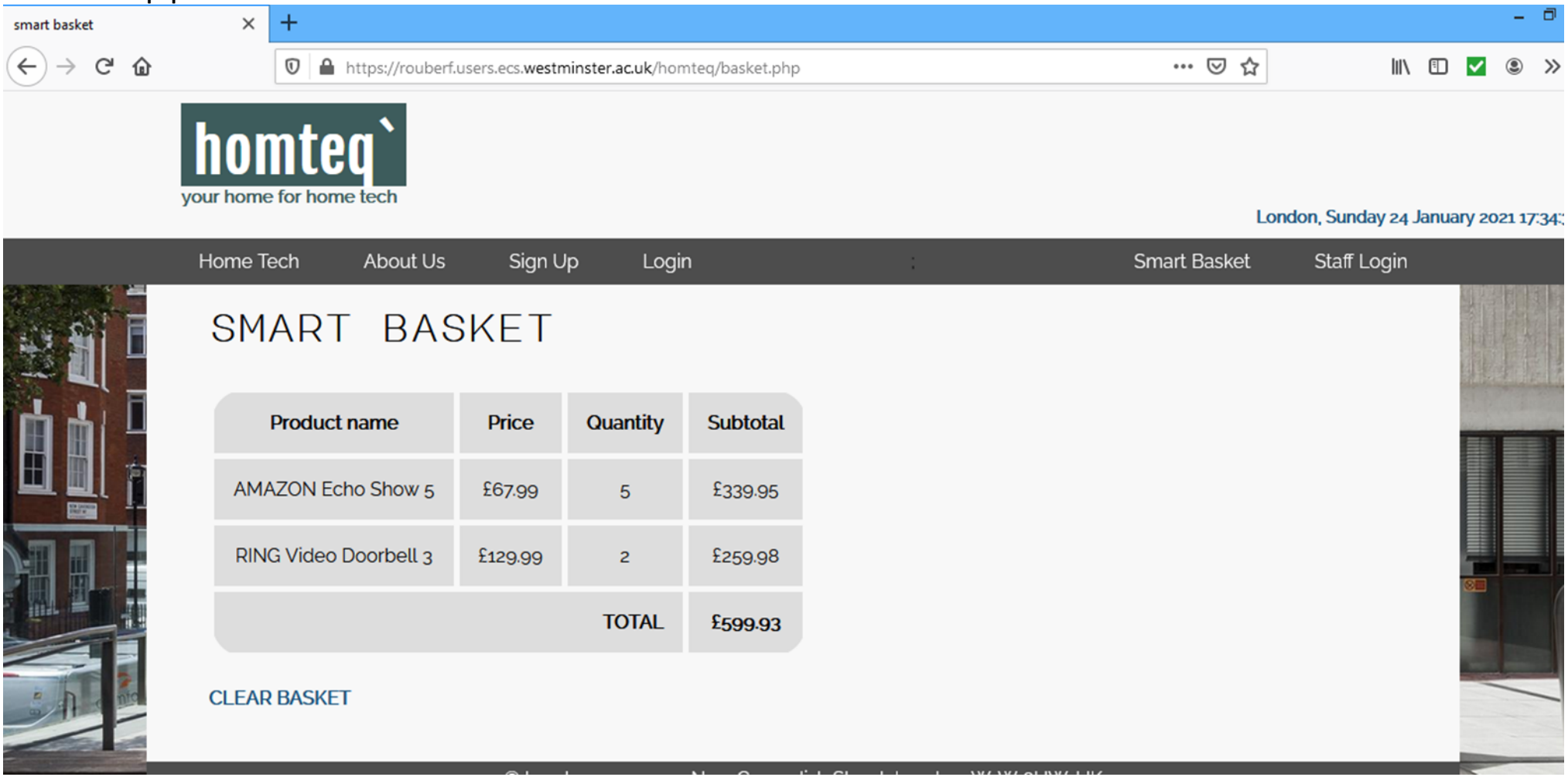
The basket has been uploaded, one more item has been added.

The details of all the products selected by the user are retrieved and displayed in the basket.

4) Then access your INDEX page and click on the “Smart Basket” link on the navigation bar to view the basket **without adding anything onto it**.

You should be able to see a display similar to this:

basket.php



The basket is unchanged, and the details of the products selected by the user are retrieved and displayed in thr basket.

TASK 05| Create a CLEARBASKET page

1) Go to your **C:\homteq** folder and, in the same folder, copy and paste your **template** file and rename the copy to **clearbasket.php.**

2) Open **basket.php** in your code editor, change the name of the page (i.e. the variable **$pagename**) to "Clear Smart Basket".

3) Right at the top under the PHP opening tag, add the session initiation and the database connection.

```
session_start();
include("db.php");
```

4) Locate the line of code that displays the name of the page using the <h4> tag.
Right under this line, write this code to clear the session array

```
unset($_SESSION['basket']);
```

5) Underneath this unset instruction, display a "Your basket has been cleared" message.

Model Answer

```
<?php
session_start();
include ("db.php");                    //include db.php file to connect to DB
$pagename="clear smart basket";        //create and populate variable called $pagename
echo "<link rel=stylesheet type=text/css href=mystylesheet.css>";
echo "<title>".$pagename."</title>";
echo "<body>";
include ("headfile.html");
echo "<h4>".$pagename."</h4>";
unset($_SESSION['basket']);
echo "<P>Your basket has been cleared!";
include ("footfile.html");
echo "</body>";
?>
```

TASK 06| Update the BASKET page so that the basket can be cleared

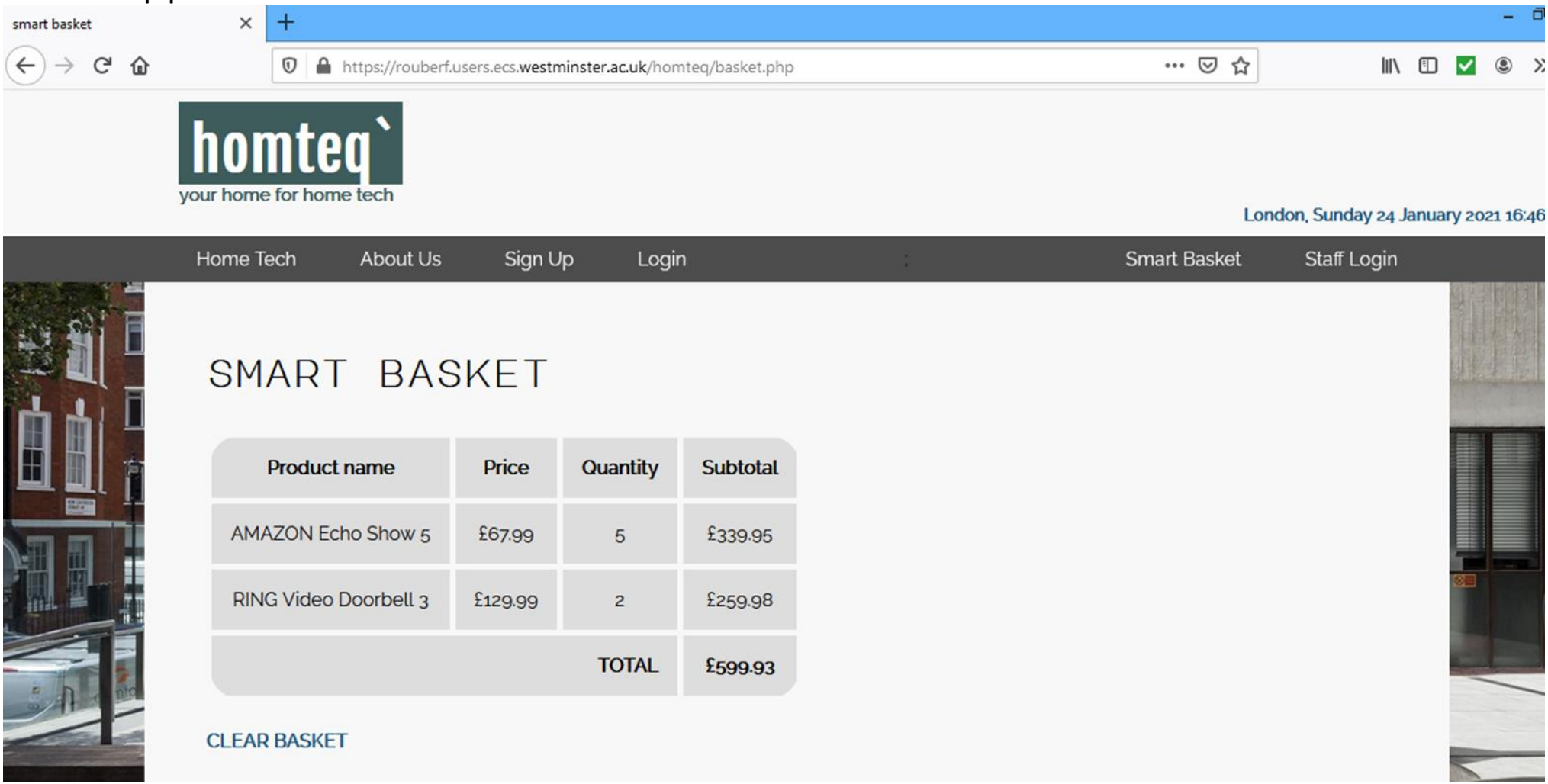
1) Open **basket.php** in your code editor, go to the bottom of the file, right before the inclusion of the foot file, and create an anchor to activate a link to **clearbasket.php** to allow the user to empty their basket.

Model Answer

```
echo "<br><p><a href='clearbasket.php'>CLEAR BASKET</a></p>";
```

2) Access the BASKET page and view the content of your current basket by clicking on the SMART BASKET link on the nav bar. You should be able to see this:

basket.php

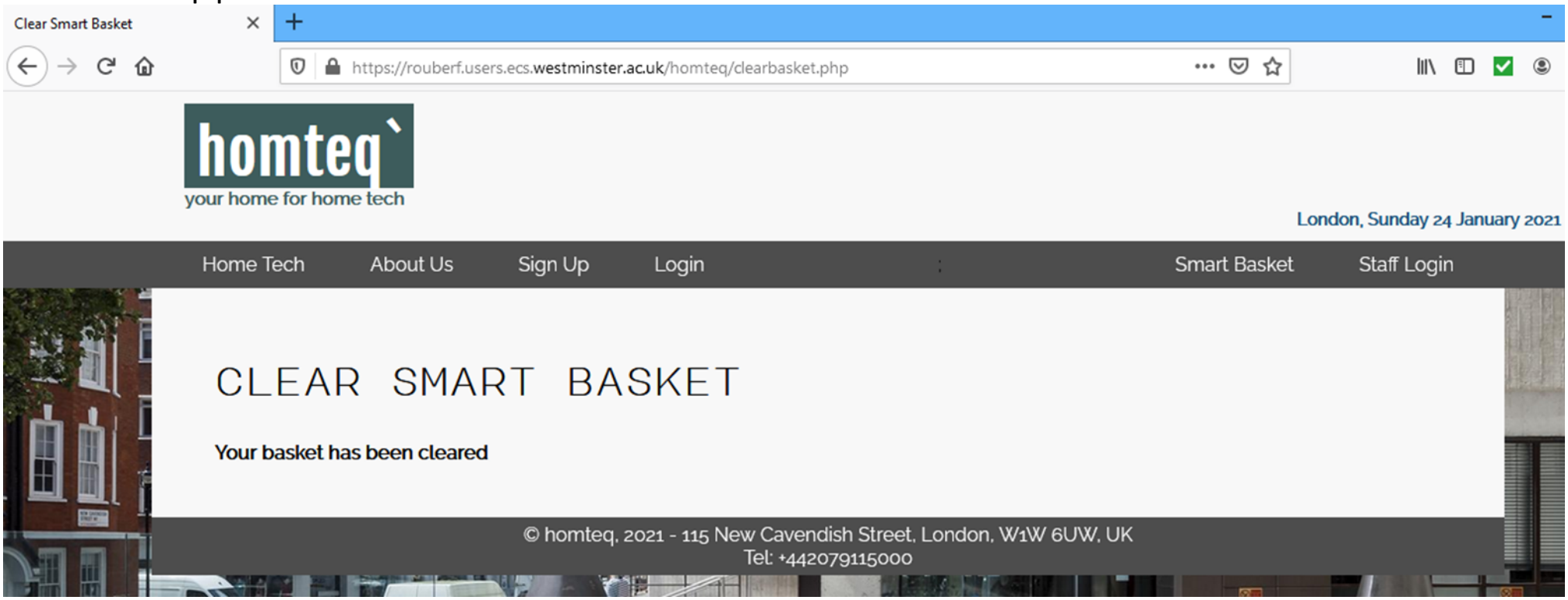


The basket is unchanged, and an anchor is displayed right under the basket to allow the user to clear it, if they so wish.

3) Click on the CLEAR BASKET anchor.

You should be able to see a display similar to this:

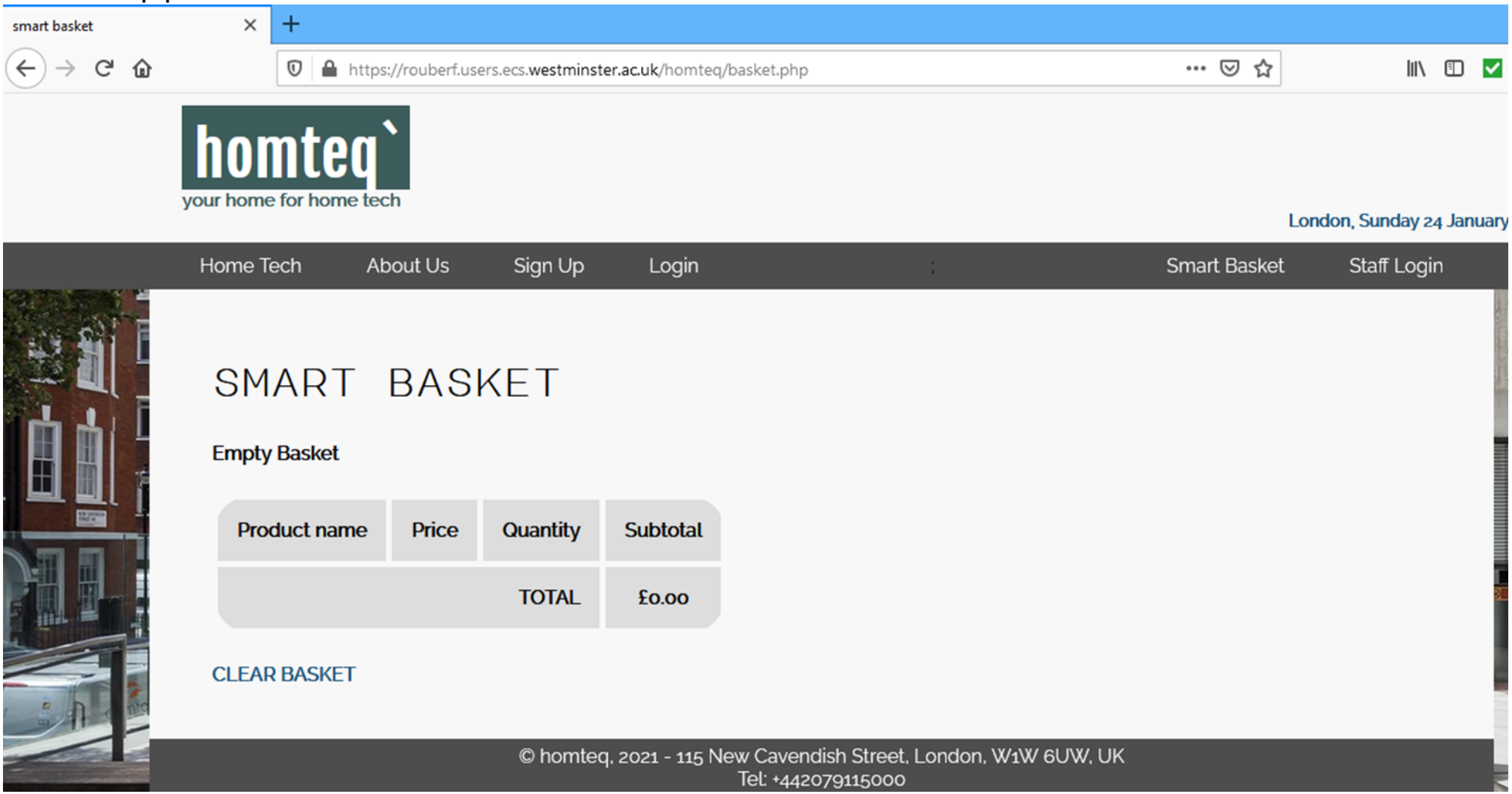
clearbasket.php



4) Access the BASKET page and view the content of your current basket again by clicking on the SMART BASKET link on the nav bar.

You should be able to see a display similar to this:

basket.php



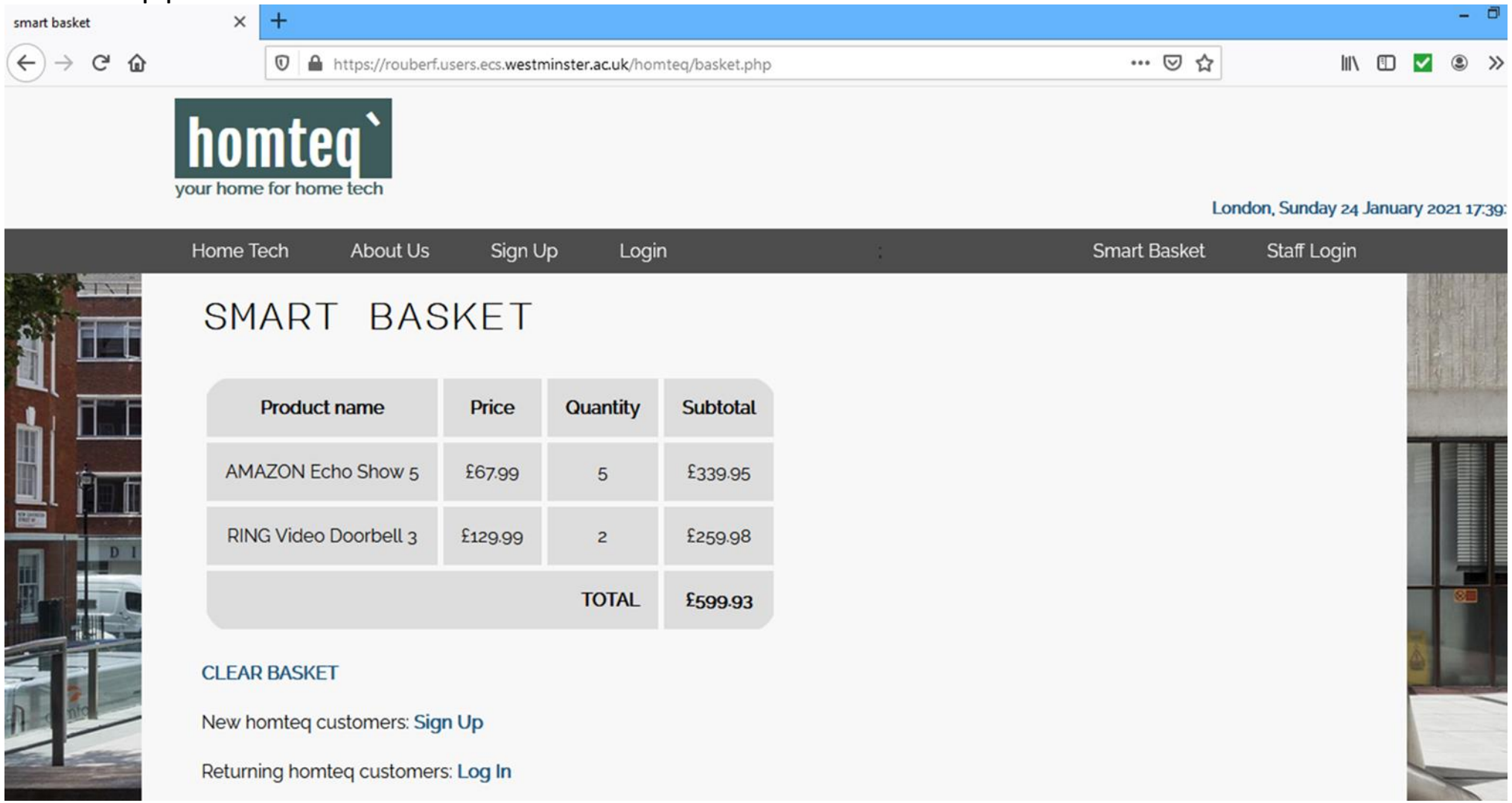
The basket is now empty.

TASK 07| Update the BASKET page so that users can either sign up or log in

1) Open **basket.php** in your code editor, go to the bottom of the file, right before the inclusion of the foot file, and create 2 anchors to activate links to **signup.php** and **login.php** to allow the user to either sign up and create an account or to log in the site and place the order.

2) Access the BASKET page and view the content of your current basket. You should be able to see a display similar to this:

basket.php



Sign up anchor to allow the user to register and create an account. Login anchor to allow the user to log into the site and place an order

Model Answer

```
echo "<br><p>New homteq customers: <a href='signup.php'>Sign up</a></p>";
echo "<p>Returning homteq customers: <a href='login.php'>Login</a></p>";
```

TASK 08| Update the BASKET page so that individual items can be removed from the basket

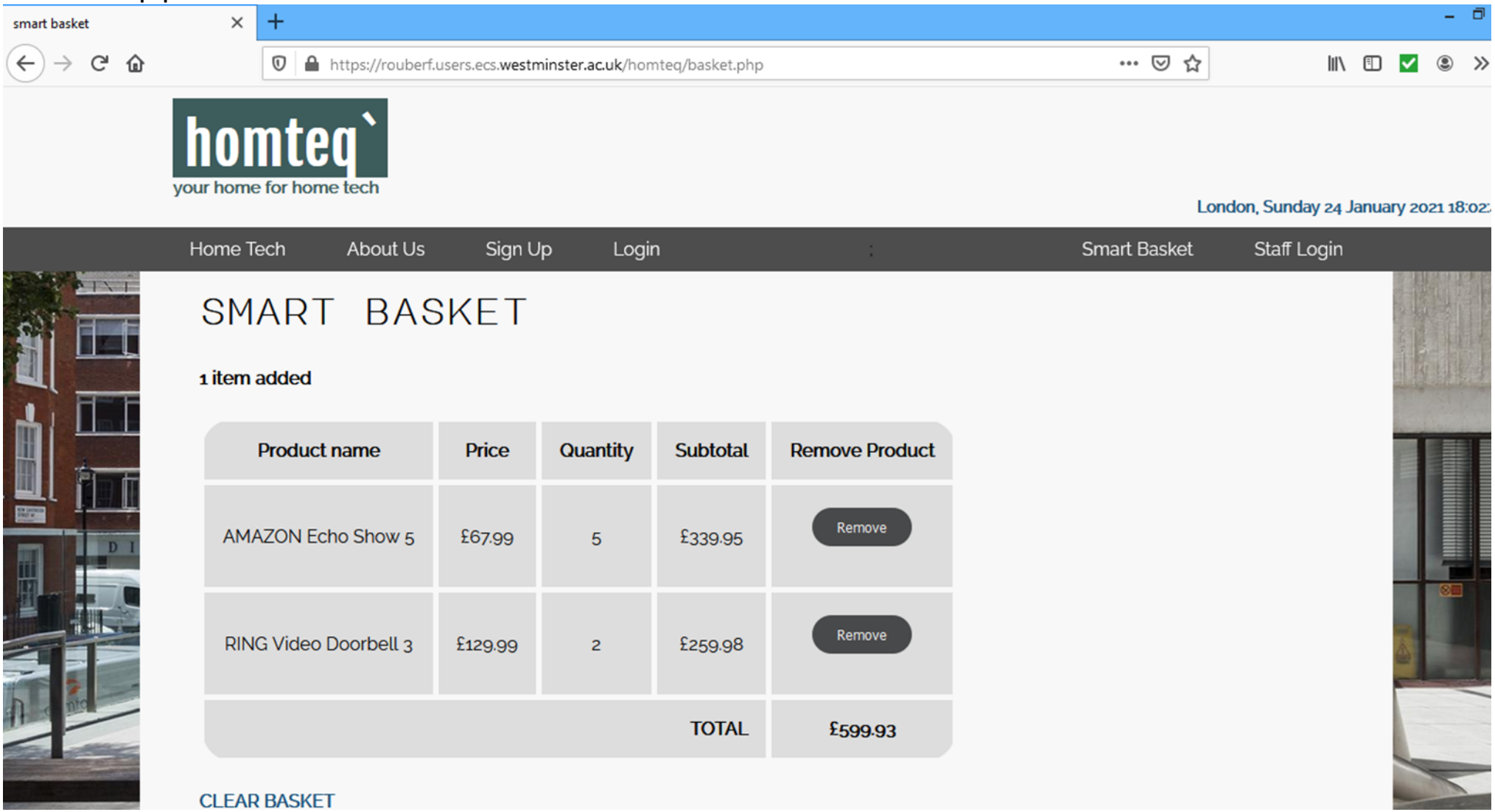
1) Modify **basket.php** to create a form that displays a REMOVE button after every item that has been placed in the basket.
The form should post values (`method = post`) to the same page (`action = basket.php`). Also use a hidden field to post through the Product Id.

Model Answer

```
echo "<form action=basket.php method=post>";
echo "<td>";
echo <input type=submit value='Remove' id='submitbtn'>";
echo "</td>";
echo "<input type=hidden name=del_prodid value=".$prodinbasketarray['prodId']."">";
echo "</form>";
```

2) Access the BASKET page and view the content of your current basket. You should be able to see a display similar to this:

basket.php



The Remove button is used to delete an item from the basket. The user is re-directed to the same page.

3) Modify **basket.php** so that to unset the cell of the session array for the id of the product selected by the user.
Right under the line that displays the name of the page using the <h4> tag, write a section of code that reflects the following pseudocode

PSEUDOCODE

```
//if the value of the product id to be deleted (which was posted through the hidden field) is set
      //capture the posted product id and assign it to a local variable $delprodid
      //unset the cell of the session for this posted product id variable
      //display a "1 item removed from the basket" message
```

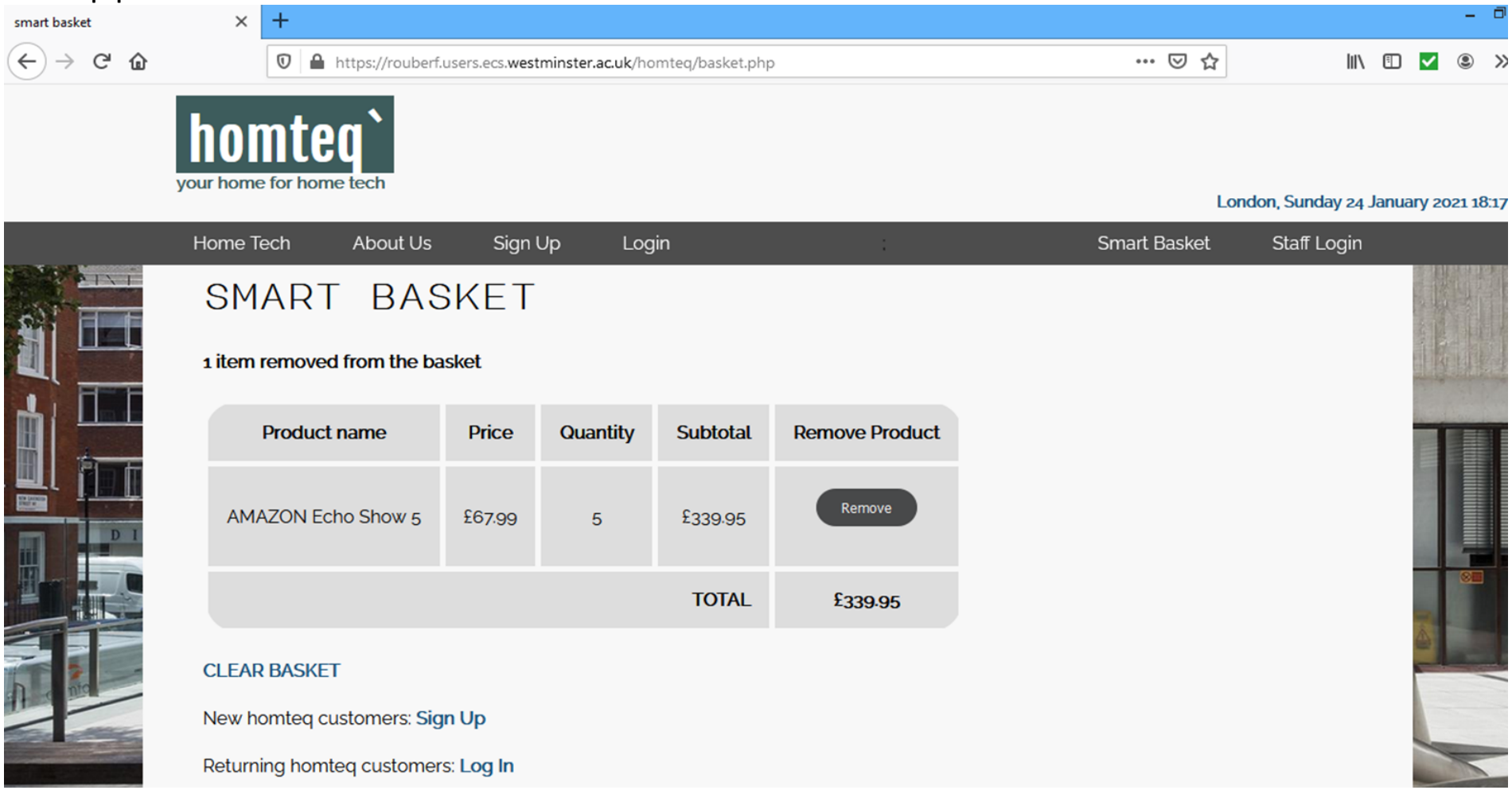
Model Answer

```
//if the value of the product id to be deleted (which was posted through the hidden field) is set
if (isset($_POST['del_prodid']))
{
      //capture the posted product id and assign it to a local variable $delprodid
      $delprodid=$_POST['del_prodid'];
      //unset the cell of the session for this posted product id variable
      unset ($_SESSION['basket'][$delprodid]);
      //display a "1 item removed from the basket" message
      echo "<p>1 item removed";
}
```

4) Access your BASKET page and click on one of the REMOVE buttons to delete an item from your basket. You should be able to see a display similar to this:

(You may want to remove the message that previously displayed “basket unchanged” as it would now be misleading).

basket.php


smart basket
https://rouberf.users.ecs.westminster.ac.uk/homteq/basket.php
homteq
your home for home tech
London, Sunday 24 January 2021 18:17
Home Tech
About Us
Sign Up
Login
Smart Basket
Staff Login
SMART BASKET
1 item removed from the basket
Product name
Price
Quantity
Subtotal
Remove Product
AMAZON Echo Show 5
£67.99
5
£339.95
Remove
TOTAL
£339.95
CLEAR BASKET
New homteq customers: Sign Up
Returning homteq customers: Log In